MILITARY SPECIFICATION

SHEARING MACHINE, THROATLESS, HAND LEVER

This specification is approved for use within the US Army Armament Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a bench type, hand operated, throatless shearing machine.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Service.
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

* 3.2 Design. The shearing machine shall be new and of the manufacturer's latest design which shall meet the requirements specified herein. The machine shall be designed to be operated on a bench or work-table and shall be capable of

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shearing, trimming, slitting and nibbling of flat metal stock, and of cutting round metal bars within the capacities specified herein. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the machine or any machine component.

* 3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the shearing machine. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in FED-STD-376.

* 3.2.2 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided means to assure adequate lubrication. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the shearing machine to remove the temporary caps or stoppers before start-up or use of the machine.

* 3.2.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

* 3.2.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.2.5 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. The hand lever (handle) shall incorporate an accident prevention device using a system of springs or similar means that shall positively hold the handle at any random position. Guards shall be removable to provide easy access to guarded parts. The shearing machine shall comply with the standards promulgated under OSHA 2206 that are applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

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* 3.4 Construction. The shearing machine shall be furnished complete so that it may be used for all operations required herein. Parts and materials shall be free of defects. Surfaces of parts shall be clean and free of extraneous material. External surfaces shall be smooth, with no sharp edges.

* 3.4.1 Castings and forgings. Castings shall be free from visible blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

* 3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

* 3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.5 Components. The shearing machine shall consist of the principal components specified in 3.5.1 through 3.5.6.

3.5.1 Base. The base shall be constructed as an integral part of the shearing machine, and shall provide a solid support for all cutting operations. The base shall be provided with at least three mounting holes or slots of not less than 0.4375 inch diameter or width.

* 3.5.2 Table. The table shall be constructed as an integral part of the shearing machine. The table shall be of a size and shape that shall assure a working area and facilities for the cutting operations specified herein.

* 3.5.3 Handle. The handle (lever) shall be fabricated of tubing or bar stock. The length of the handle shall be commensurate with the force required for each shear application (see 3.6). One or more varying length handles may be provided at the manufacturer's option.

3.5.4 Blades and rod dies. The blades and rod dies shall be of hardened tool steel (see 3.5.4.1) having not less than one percent carbon content. Blades shall be provided with adjustments for wear. The blades and rod dies shall be designed to be interchanged. The length of the blade shall be not less than 3.5 inches.

* 3.5.4.1 Heat treatment. Heat treatment of the blades and rod dies shall conform to the best commercial practice. Blades and rod dies shall have a hardness of 45 to 60 on the Rockwell C scale.

3.5.5 Holddown. The shearing machine shall be provided with a holddown, or a similar device, for holding the metal securely during the cutting operation.

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The holddown shall operate smoothly and shall be adjustable for all sizes of metal within the capacity of the machine (see 3.6).

3.5.6 Actuating mechanism. The actuating mechanism shall be of a mechanical design that shall provide for sufficient transfer of power from the handle to the cutter.

* 3.6 Capacity. The shearing machine shall be capable of slitting, shearing, inside and outside nibbling of 0.187 inch mild steel stock, trimming of 0.25 inch mild steel bar stock, and rod cutting of 0.187 through 0.375 inch diameter. The actuating force applied to the end of the handle shall not exceed 105 pounds for rod cutting, and 185 pounds for all other cutting operations.

3.7 Dimensions. The overall dimensions of the shearing machine shall not exceed 16 inches in height (without handle), 11 inches in width, and 13 inches in depth.

* 3.8 Weight. The total weight of the shearing machine shall not exceed 60 pounds.

* 3.9 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.10 Product identification. The shearing machine shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

* 3.11 Workmanship. Standards of workmanship shall assure that the shearing machine shall have the stability, strength, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the

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contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (d) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Capacity test (see 4.6.1).
(c) Hardness test (see 4.6.2).
(d) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination	II	1.0
Capacity test	II	1.0
Hardness test	II	1.0
Packaging inspection	S-4	4.0

* 4.5 Product examination. Visually, dimensionally, and manually examine the shearing machine to determine conformance with the requirements of 3.2 through 3.5.4, 3.5.5 and 3.5.6, and 3.7 through 3.11. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the shearing machine has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection.

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4.6 Tests.

4.6.1 Capacity test. The shearing machine shall be tested to determine compliance with 3.6. One piece of material 12 inches long, of the size specified in 3.6, shall be provided for each operation. The cutting force required to cut the material shall be measured by using standard measuring test equipment at the end of the handle. The pull or force required to cut the material shall not exceed the actuating force specified in 3.6. The shearing machine shall perform all the cutting requirements in 3.6 without damage to the actuator, handle, blades, rod dies, or any other part of the machine.

4.6.2 Hardness test. Hardness tests shall be performed on the shearing blades and rod dies to determine compliance with 3.5.4.1.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation, Levels A or B packing, and military marking shall be in accordance with MIL-M-18058.

* 5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The shearing machine covered by this specification is intended to be used in maintenance repair shops for shearing, slitting, inside and outside nibbling, trimming flat metal stock, and cutting round bar stock.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Exceptions and additional safety/health requirements when required (see 3.2.5 and 6.4).
(d) Painting and finishing if different (see 3.9).
(e) Product identification if different (see 3.10).
(f) First article inspection if different (see 4.3).
(g) Quality conformance inspection if different (see 4.4).
(h) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

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* 6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

* 6.5 Standard lubricants. The shearing machine manufacturer should specify and use to the maximum extent possible those lubricants (see 3.2.2) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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